

LIU HENGZHI

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1. EDUCATION

National University of Singapore

2026 – Present

Bachelor of Computing in Computer Science

Singapore

Temasek Polytechnic

2023 – 2026

Diploma in Biomedical Engineering with Merit (GPA: 3.91/4.00)

Singapore

- Polytechnic Scholarship (Engineering); Course Bronze Medalist; Director's List
- Focused on embedded systems, hardware–software co–design, and applied engineering research

2. SKILLS

Programming Languages: Python, C++, C, Swift

Systems & Backend: Embedded Systems, Real–time Systems (FreeRTOS), System Design Basics, Firmware Development

Hardware & Edge Systems: ESP32, Arduino, PCB Design (LCEDA), Sensor Integration, Edge Computing Concepts

Tools: Git/GitHub, Linux (basic), CAD (Fusion 360 / NX / Creo), Adobe Suite

3. WORK EXPERIENCE

National University of Singapore (iHealthtech)

Mar 2025 – Jul 2025

Research Assistant Intern

Singapore

- Conducted cell culture and biological experiments following established laboratory protocols and safety standards.
- Operated microfluidic and particle characterization instruments for experimental studies and data collection.
- Analyzed experimental datasets and prepared technical reports to support ongoing research activities.
- Assisted in laboratory operations including procurement of research materials, inventory management, and coordination of experimental workflows.
- Maintained accurate documentation of experimental procedures and results to ensure research reproducibility.

4. TECHNICAL PROJECTS

Smart Mahjong Play-Board

2026

Independent Systems Developer

C++, ESP32, PCB Design, Embedded Firmware

- Designed and built an ESP32-based smart Mahjong play-board prototype with custom PCB integration.
- Developed embedded firmware and hardware interfaces for real-time game interaction.

Experiential Research Programme

2024

Research Student

- Investigated titanium electroplating techniques for fuel-cell applications through laboratory experimentation and analysis.

5. UNIVERSITY ENGAGEMENT

- Incoming Undergraduate Student at School of Computing, National University of Singapore.
- Active engagement in computing community frameworks and prospective engineering alignment programs.

6. CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing)

2024

- 1st Place (Qualifying Round); demonstrated advanced CAD modeling, additive manufacturing optimization, and engineering precision design

Electronic Design & Robotics Competitions

2024

- Developed embedded logic systems and real-time control algorithms for robotics and circuit design challenges

7. ADDITIONAL INFORMATION

- **Leadership:** Temasek LEAD & ENGenius Excellence Programs alumnus
- **Languages:** English (Professional), Mandarin (Native)
- **Interests:** Distributed systems, AI infrastructure, embedded systems, scientific computing